

FOURTH SEMESTER | 2024 SCHEME | GAMAT401

Mathematics for Information Science – 4

MODEL PAPER — ANSWER KEY



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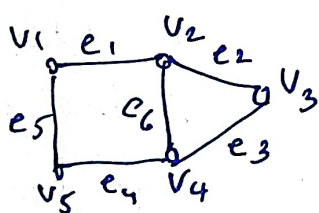
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(VTJCT)

- 1) Since each edge has two end vertices it contributes two to the sum of degrees. i.e., when the degrees of vertices are counted each edge is counted twice including loop.
 $\therefore \sum d(v_i) = 2e.$

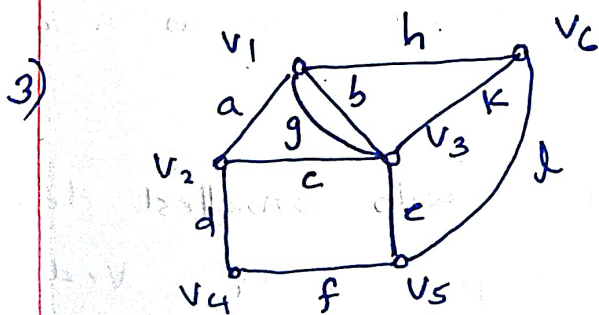
- 2) Walk: Is a finite alternating sequence of vertices and edges beginning & ending with vertices.
 Path: An open walk in which no vertex appears more than once.
 Circuit: A closed path is called circuit.



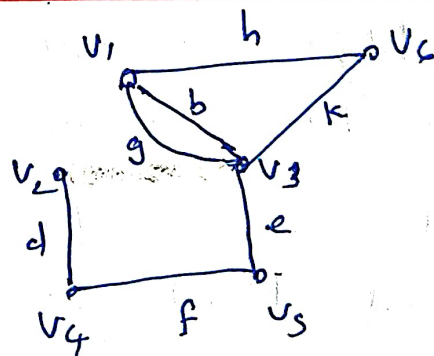
Walk: $v_1, e_1, v_2, e_6, v_4, e_3, v_3, e_2, v_2$

Path: $v_1, e_1, v_2, e_2, v_3, e_3, v_4$

Circuit: $v_1, e_1, v_2, e_6, v_4, e_4, v_5, e_5, v_1$



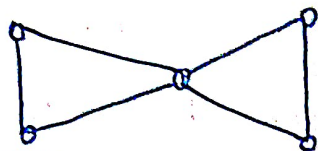
$G_1 \cup G_2$



$G_1 \oplus G_2$

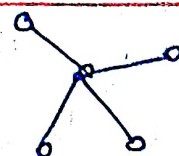
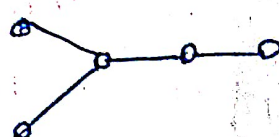
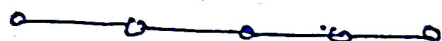
4. A graph which contains a closed walk which pass through every edge exactly once is called an Euler graph.

Eg:



Euler but not Hamiltonian

5



6

$$n \leq 2^0 + 2^1 + 2^2 + 2^3 = 1 + 2 + 4 + 8 = 15$$

Maximum height, $\max h_{\max} = \frac{n-1}{2} = \frac{15-1}{2} = \frac{14}{2} = 7$

7

Let G be a connected graph & T be a sp. tree. Let S be an arbitrary cut-set in G . We have to prove that S must contain at least one branch of T .

So assume the converse. Assume that the cut set S and spanning tree T has no edge in common. The removal of the edge in the cut set wouldn't disconnect the graph. This is a contradiction. Hence the theorem.

8

Let v_i be the vertex with smallest degree in G . Let $d(v_i)$ be the degree of v_i . Vertex v_i can be separated from G by removing the $d(v_i)$ edges incident on vertex v_i .

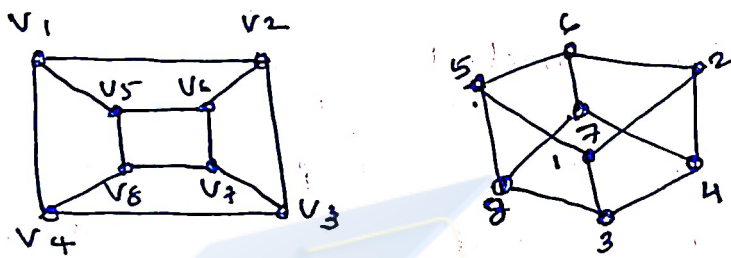
Hence the Theorem.

9 a)

The two graphs are isomorphic.

Since there is a one-to-one correspondence between the vertices & edges of the graphs such that the incidence relationship is preserved.

The correspondence can be given as follows



$v_1 \rightarrow 1$	$v_5 \rightarrow 5$
$v_2 \rightarrow 2$	$v_6 \rightarrow 6$
$v_3 \rightarrow 3$	$v_7 \rightarrow 7$
$v_4 \rightarrow 4$	$v_8 \rightarrow 8$

The two graphs have same number of vertices, same number of edges & same number of vertices with same degree.

9 b)

Assume that the graph is not connected.

Then it has at least two components.

To get the maximum number of edges in a disconnected graph we make one component with $(n-1)$ vertices and other component with one vertex.

Maximum no. of edges in the component with $(n-1)$ vertices is $\frac{(n-1)(n-2)}{2}$

The single isolated vertex contributes 0 edges.

So total maximum edges in a disconnected graph is $\frac{(n-1)(n-2)}{2}$.

i.e., a disconnected graph can have at most $\frac{(n-1)(n-2)}{2}$ edges.

So if a graph has more than $\frac{(n-1)(n-2)}{2}$ edges it must be connected.

10 a) Complete graph is a simple graph in which each pair of distinct vertices are joined by an edge.

Complete bipartite graph G is a graph whose vertex set can be partitioned into two non-empty sets X & Y such that $X \cup Y = V$, $X \cap Y = \emptyset$ and every vertex in X is connected to every other vertex in Y through an edge.

K_2 : $v_1 \text{ --- } v_2$

$K_{1,1}$:

complete & complete bipartite graph.

10 b) Let v_1 & v_2 be the two vertices of odd degree in a graph G . If G is a connected graph, then surely there exists a path between every vertex and hence between v_1 & v_2 .

Let G be a disconnected graph. If v_1 & v_2

belong to the same component of G , then they are joined by a path.

Now assume that v_1 & v_2 are in two different components. Then each component consists of a single vertex of odd degree which contradicts the theorem that there is even number of odd vertices in a graph.

Hence our assumption of v_1 & v_2 belong to two different components is wrong.

i.e., v_1 & v_2 belongs to the same component and hence joined by a path.

11 a) A directed graph G consists of a vertex set $V = \{v_1, v_2, \dots\}$ and an edge set $E = \{e_1, e_2, \dots\}$ where every edge e_k is represented by some ordered pair of vertices (v_i, v_j) .

Digraphs in which for every edge (a, b) there is also an edge (b, a) is called a symmetric digraph.

Digraphs that have at most one directed edge between a pair of vertices is called an asymmetric graph.

11 b) This is the seating arrangement problem which can be solved as follows.

Represent a member by a vertex

possibility of sitting next to another member by an edge between them.

Since every member is allowed to sit next to any other member G is a complete graph with 9 vertices.

9 is the number of people to be seated around the table.

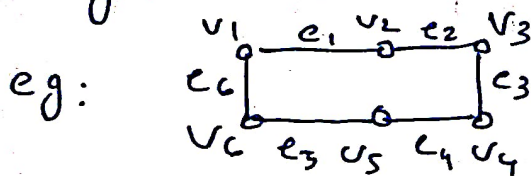
Every seating arrangement around the table is clearly a Hamiltonian circuit.

By the theorem, the number of edge disjoint Hamiltonian cycle is $\frac{n-1}{2} = \frac{9-1}{2} = \frac{8}{2} = \underline{4}$

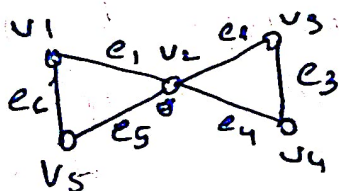
$\therefore$ 4 such arrangement exists among 9 people.

12 a) Hamiltonian circuit is defined as a closed walk that traverses every vertex of G exactly once, except the starting vertex at which the walk terminates.

Hamiltonian path: is a path which contains every vertex of G exactly once.



$v_1 e_1 v_2 e_2 v_3 e_3 v_4 e_4 v_5 e_5 v_6 e_6 v_1$



$v_1 e_6 v_5 e_5 v_2 e_2 v_3 e_3 v_4$

12b)

Suppose that G can be decomposed into circuits. Since the degree of every vertex in a circuit is two, the degree of every vertex in G is even. Hence G is an Euler Graph.

Conversely, let G be an Euler graph. Consider a vertex v_1 . There are at least two edges incident with v_1 .

Let one of these edges be between v_1 & v_2 . Since v_2 is also of even degree it must have at least another edge between v_2 & v_3 .

Proceeding like this gradually we arrive at a vertex that has previously been traversed thus forming a circuit C .

Let us remove C from G . All vertices in the remaining graph must also be of even degree.

From the remaining graph remove another circuit in exactly same way as we removed C from G .

Continue this process until no edges are left. Hence an Euler graph can be decomposed into circuits.

13)

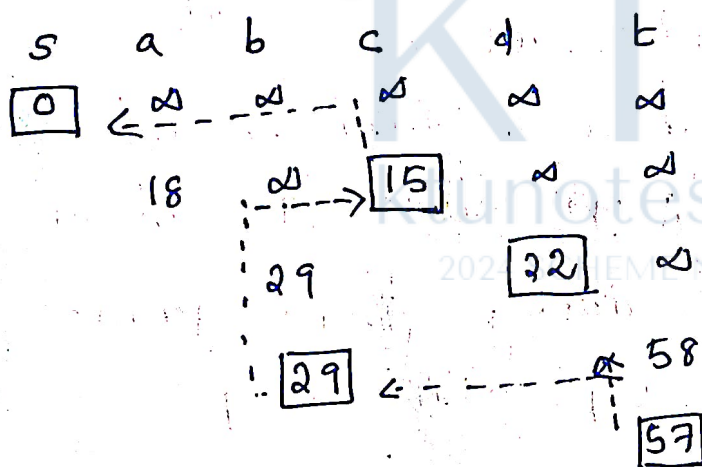
Dijkstra's Algorithm:

Step I: Assign a permanent label '0' to the starting vertex s and a temporary label ∞ to the remaining $(n-1)$ vertices.

Step II: Every vertex 'j' that is not yet permanently labelled, gets a new temporary label given by $\min \{ \text{old label of } j, \text{ old label of } i + d_{ij} \}$

Step III: The smallest value among the temporary labelled vertices is found, and that becomes the permanent label of the corresponding vertex. In case of a tie select any one of them.

Step IV: Step II & III are repeated alternately until the destination vertex t gets a permanent label.



$s \rightarrow c \rightarrow b \rightarrow t$, is the shortest path with distance 57

- 14) Let us prove the theorem by induction on number of vertices. for $n=1, 2, \& 3$ theorem is true.
 $n=1$ • 0 edge, $n=2$ • 1 edge, $n=3$ • 3 edge
 Assume that theorem holds for all trees with fewer than n vertices. Now consider a tree with n vertices.

In T let e_k be an edge with end vertices v_i & v_j . Since by theorem there is no other path between v_i & v_j except e_k . Hence the deletion of e_k from T will disconnect the graph. Furthermore $T - e_k$ consists of exactly two components and each of them components is a tree.

Both these trees T_1 & T_2 have fewer than n vertices each and therefore by the induction hypothesis each contain one edge less than number of vertices. Thus $T - e_k$ consists of $n-2$ edges.

Hence T has exactly $n-1$ edges.

15

Cutset is the minimal set of edges whose removal will disconnect the graph.

Let S be a cutset in G . Then by definition of cut set, the removal of S partitions the vertices of G into two subsets V_1 & V_2 .

Consider a circuit C in G . If all vertices of C are entirely in the set V_1 (or V_2) the number of edges common to S & C are zero. (Clearly even)

If on the other hand some vertices in C are in V_1 and some in V_2 . We travel back & forth between two sets V_1 & V_2 to get a circuit. Because of the closed nature of a circuit, the number of edges we traverse between V_1 & V_2 must be even. Since S is a cutset every edge in S has one end in V_1 and the other end in V_2 . and no other edge in G has this property.

$\therefore$ The no. of edges common to S & C are even.

16) (i) Incidence Matrix:

	a	b	c	d	e	f	g	h
v_1	0	0	0	1	0	1	0	0
v_2	0	0	0	0	1	1	1	1
v_3	0	0	0	0	0	0	0	1
v_4	1	1	1	0	1	0	0	0
v_5	0	0	1	1	0	0	1	0
v_6	1	1	0	0	0	0	0	0

(ii) Adjacency Matrix:

	v_1	v_2	v_3	v_4	v_5	v_6
v_1	0	1	0	0	1	0
v_2	1	0	1	1	1	0
v_3	0	1	0	0	0	0
v_4	0	1	0	0	1	2
v_5	1	1	0	1	0	0
v_6	0	0	0	2	0	0

(iii) paths $\rightarrow$ 1- $\{d, c\}$

2- $\{d, g, e\}$

3- $\{f, e\}$

4- $\{f, g, c\}$

$P(v_1, v_4) =$

	a	b	c	d	e	f	g	h
1	0	0	1	1	0	0	0	0
2	0	0	0	1	1	0	1	0
3	0	0	0	0	1	1	0	0
4	0	0	1	0	0	1	1	0

(iv) Circuits: 1 $\rightarrow$ $\{g, c, e\}$

2 $\rightarrow$ $\{f, d, g\}$

3 $\rightarrow$ $\{e, c, d, f\}$

4 $\rightarrow$ $\{a, b\}$

	a	b	c	d	e	f	g	h
1	0	0	1	0	1	0	1	0
2	0	0	0	1	0	1	1	0
3	0	0	1	1	1	1	0	0
4	1	1	0	0	0	0	0	0

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